Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/23/2006
PI Name:	DeLucas, Lawrence J. O.D., Ph.D.		
Project Title:	Protein Crystal Growth in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Other		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Lawrence.delucas@aero.org	**Fax:**	FY
PI Organization Type:	NON-PROFIT	**Phone:**	3103365000
Organization Name:	The Aerospace Corporation		
PI Address 1:	Center for Biophysical Sciences and Engineering		
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PI Web Page:			
City:	El	**State:**	CA
Zip Code:	90245	**Congressional District:**	33
Comments:	NOTE: PI retired in 2017 from the University of Alabama, Birmingham, and is now Principal Scientist at The Aerospace Corporation, El Segundo, CA (Ed., 5/22/2018)		
Project Type:	Flight	**Solicitation / Funding Source:**	OTHER
Start Date:	04/01/1994	**End Date:**	12/01/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Karr, Laurel	**Contact Phone:**	(256) 544-7817
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAS8-40189		
Performance Goal No.:			
Performance Goal Text:			

Two laboratory based systems have been constructed to demonstrate methods which will allow for dynamic control of protein crystal growth. The technologies developed in these systems were to be incorporated into future flight hardware for use in microgravity studies. Unfortunately, the temperature-induced crystallization system was discontinued due to insufficient funds in the NASA budget. The first system uses a precisely controlled vapor diffusion approach to monitor and control protein crystal growth. The vapor crystallization system utilizes a humidity sensor and various interfaces under computer control to establish different evaporation rates in many different growth solutions simultaneously. A static laser light scattering sensor can be used to detect aggregation events and trigger a change in the evaporation rate for a growth solution. The system demonstrates significant advances in the ability of researchers to gain control of the protein crystal growth process and will provide tremendous opportunities for both terrestrial and microgravity research.

Task Description:	The objectives of this research are the following: 1) To develop improved protein crystal growth flight hardware to produce large, high quality crystals in microgravity for use in the determination of protein and other macromolecular three-dimensional structures; and, 2) To understand the thermodynamics and kinetics of macromolecular crystal growth. Large, high quality crystals are necessary for the determination of the molecular structure of macromolecules by x-ray diffraction analysis. This structural information has applications in all the biological sciences, especially medicine, drug design and agriculture. Previous research has demonstrated that macromolecular crystals growth in microgravity are frequently larger and more perfectly formed than their earth-grown counterparts. This improvement in size and quality translates into x-ray diffraction data of higher resolution and intensity, yielding better structural information about the molecule.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this reporting period.
Bibliography Type:	Description: (Last Updated: 06/29/2023)